

JONAS SPINNER

CURRICULUM VITAE

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RESEARCH INTERESTS

- ML tools for the LHC
 - Geometric deep learning
 - Generative modelling
- Data analysis: simulation, reconstruction, inference
Lorentz-equivariant graph networks and transformers
Flow matching, autoregressive methods, generation on manifolds

EXPERIENCE

Feodor Lynen Fellow IPPP, Durham University 10/2025 – Exp. 10/2027

- Topics: Physics-informed ML for high-energy physics

EDUCATION

PhD in Physics Heidelberg University 10/2022 – 09/2025

- Thesis: Symmetry-aware networks for the LHC (advisor: Tilman Plehn)
- Topics: ML for the LHC, generative modelling, Lorentz-equivariant networks

MSc in Physics Karlsruhe Institute for Technology 1.1* 10/2020 – 09/2022

- Thesis: New Light Particles in Astrophysics, Cosmology and at Colliders (advisor: Ulrich Nierste)
- Visited IPPP, Durham University for 6 months during the MSc Thesis

BSc in Physics Karlsruhe Institute for Technology 1.1* 10/2017 – 09/2020

- Thesis: Dimension 7 operators in Tritium Beta Decay (advisor: Ulrich Nierste)

Abitur Robert-Gerwig Gymnasium Hausach 1.0* 06/2017

*German grading scale: from 1.0 (best) to 6.0 (worst)

TEACHING

Lecture on Machine Learning for Particle Physics at IPPP, Durham University 04/2026

- Four 120-minute lectures on machine learning for PhD students

Co-supervision MSc thesis Sebastian Pitz 08/2024 – 07/2025

- Topic: Jet Tagging with Learned Lorentz Equivariant Frames

Lecturer at Erum Datahub Active Training Course on Advanced Deep Learning 05/2024

- Delivered a 90-minute lecture on Transformers for 50 students

Head teaching assistant for MSc-level course Machine Learning and Physics 10/2023 – 03/2024

- Organization of exercises and exams for 120 students

Co-supervision BSc thesis Nathanael Ediger 06/2023-10/2023

- Topic: LHC Event Generation with JetGPT - From variable orderings to joint training

Teaching assistant for 7 one-term courses in Theoretical Physics 04/2019 – 09/2023

- Discussions and marking for groups of 10-20 students

SKILLS

Technical Python, PyTorch, git, $\text{\LaTeX}$, SLURM/HPC, Wolfram Mathematica
Languages German (native), English (fluent)

AWARDS

- Feodor Lynen Research Fellowship (2-year fellowship with Humboldt foundation)
- Prestigious German Studienstiftung scholarship (top 0.5% of students nationwide)
- Deutschlandstipendium scholarship (awarded to top-performing students across Germany)

PROFESSIONAL SERVICE

Co-organizer institute seminar IPPP Durham	Since 05/2026
SciPost reviewer	Since 2025
Co-organizer CRC Young Scientists Meeting	07/2025
NeurIPS reviewer	2025, 2026
Co-organizer ITP Heidelberg seminar “Modern Numerics for Theoretical Physics”	10/2024 – 07/2025

WORKSHOPS AND CONFERENCE TALKS

For the full titles and slides of my talks, see github.com/spinjo/slides.

Corfu Workshop	Lorentz-equivariant networks	Corfu, GR	04/2026
MG Workshop	Lorentz-equivariant transformers	Heidelberg, DE	02/2026
NeurIPS	Lorentz Local Canonicalization	San Diego, US	12/2025
ACAT	Extrapolating jet radiation with transformers	Hamburg, DE	09/2025
MIAPbP Workshop	Lorentz-equivariant transformers	Munich, DE	09/2025
IAIFI Workshop	Lorentz Local Canonicalization	Boston, US	08/2025
IRN Terascale	Lorentz-Equivariant Geometric Algebra Transformer	Strasbourg, FR	05/2025
NeurIPS	Lorentz-Equivariant Geometric Algebra Transformer	Vancouver, CA	12/2024
ML4Jets	Lorentz-equivariant Geometric Algebra Transformer	Paris, FR	11/2024
PHYSTAT Workshop	Lorentz-Equivariant Geometric Algebra Transformer	London, UK	09/2024
EuCAIFCON	JetDiffusion vs JetGPT	Amsterdam, NL	04/2024
ML4Jets	JetDiffusion vs JetGPT	Hamburg, DE	11/2023
IRN Terascale	JetDiffusion vs JetGPT	Marseille, FR	10/2023
YTF	Axion-Higgs Portal	Durham, UK	12/2022

SEMINAR TALKS

CP3	Lorentz-equivariant networks	Brussels, BE	02/2026
IPPP	Lorentz-equivariant networks	Durham, UK	07/2025
COMETA	Lorentz-equivariant Geometric Algebra Transformer	Remote	01/2025
ATLAS Flavor group	Lorentz-equivariant Geometric Algebra Transformer	Remote	11/2024

PUBLICATIONS AND PREPRINTS

- [1] J. Spinner and J. Shergold, *Neural Boltzmann Equations* (2026), [arXiv:2608.23022](https://arxiv.org/abs/2608.23022)
- [2] L. Favaro, T. Plehn, H. Qu and J. Spinner, *Virtues and Vices of Equivariant Transformers* (2026), [arXiv:2608.02735](https://arxiv.org/abs/2608.02735)
- [3] A. Petitjean, T. Plehn, J. Spinner and U. Köthe, *Economical Jet Taggers – Equivariant, Slim, and Quantized* (2025), [arXiv:2512.17011](https://arxiv.org/abs/2512.17011)
- [4] A. Petitjean, A. Butter, K. Greif, S. Palacios Schweitzer, T. Plehn, J. Spinner and D. Whiteson, *Generative Unfolding of Jets and Their Substructure* (2025), [arXiv:2510.19906](https://arxiv.org/abs/2510.19906)
- [5] H. Bahl, S. Diefenbacher, N. Elmer, T. Plehn and J. Spinner, *Forecasting Generative Amplification*, SciPost Phys. **20**, 150 (2026), doi:[10.21468/SciPostPhys.20.5.150](https://doi.org/10.21468/SciPostPhys.20.5.150), [arXiv:2509.08048](https://arxiv.org/abs/2509.08048)
- [6] L. Favaro, G. Gerhartz, F. A. Hamprecht, P. Lippmann, S. Pitz, T. Plehn, H. Qu and J. Spinner, *Lorentz-Equivariance without Limitations* (2025), [arXiv:2508.14898](https://arxiv.org/abs/2508.14898)
- [7] C. A. Manzari, J. Martin Camalich, J. Spinner and R. Ziegler, *The SN 1987A Cooling Bound on Dark Matter Absorption in Electron Targets* (2025), [arXiv:2508.00725](https://arxiv.org/abs/2508.00725)
- [8] J. Spinner, L. Favaro, P. Lippmann, S. Pitz, G. Gerhartz, T. Plehn and F. A. Hamprecht, *Lorentz Local Canonicalization: How to Make Any Network Lorentz-Equivariant*, In *39th conference on Neural Information Processing Systems* (2025), [arXiv:2505.20280](https://arxiv.org/abs/2505.20280)
- [9] A. Butter, F. Charton, J. M. Villadamigo, A. Ore, T. Plehn and J. Spinner, *Extrapolating Jet Radiation with Autoregressive Transformers*, SciPost Phys. **20**, 004 (2026), doi:[10.21468/SciPostPhys.20.1.004](https://doi.org/10.21468/SciPostPhys.20.1.004), [arXiv:2412.12074](https://arxiv.org/abs/2412.12074)

- [10] J. Brehmer, V. Bresó, P. de Haan, T. Plehn, H. Qu, J. Spinner and J. Thaler, *A Lorentz-equivariant transformer for all of the LHC*, SciPost Phys. **19**, 108 (2025), doi:10.21468/SciPostPhys.19.4.108, arXiv:2411.00446
- [11] J. Spinner, V. Bresó, P. de Haan, T. Plehn, J. Thaler and J. Brehmer, *Lorentz-Equivariant Geometric Algebra Transformers for High-Energy Physics*, In 38th conference on Neural Information Processing Systems (2024), arXiv:2405.14806
- [12] C. A. Manzari, J. Martin Camalich, J. Spinner and R. Ziegler, *Supernova limits on muonic dark forces*, Phys. Rev. D **108**, 103020 (2023), doi:10.1103/PhysRevD.108.103020, arXiv:2307.03143
- [13] A. Butter, N. Huetsch, S. Palacios Schweitzer, T. Plehn, P. Sorrenson and J. Spinner, *Jet diffusion versus JetGPT – Modern networks for the LHC*, SciPost Phys. Core **8**, 026 (2025), doi:10.21468/SciPostPhysCore.8.1.026, arXiv:2305.10475
- [14] M. Bauer, G. Rostagni and J. Spinner, *Axion-Higgs portal*, Phys. Rev. D **107**, 015007 (2023), doi:10.1103/PhysRevD.107.015007, arXiv:2207.05762